

PROJECT PLAN

Enterprise Learning Platform with Skill and Career Guidance System

A structured plan for developing an AI-assisted learning and career-readiness platform.

PROJECT DETAILS	INFORMATION
Project Title	Enterprise Learning Platform with Skill and Career Guidance System
Project Duration	12 Weeks
Start Date	29 June 2026
Frontend	React, Vite and Tailwind CSS
Backend	Spring Boot, Spring Security and JPA/Hibernate
Database	MySQL
AI	Google Gemini API
Architecture	Client-server architecture using REST APIs
Project Nature	Full-stack learning and career-guidance platform

TEAM

Role / Area	Member
Frontend	Kartik Saini, Harshit Raj
Backend	S. Kavipriya
Database	Gayathri
AI Integration	Kartik Saini
Mentor	Shakthi Gopala Krishnan

Prepared for academic project submission.

1. Project Overview

The Enterprise Learning Platform with Skill and Career Guidance System is a full-stack learning and career-guidance platform. The system is designed to take a student through the complete journey from course enrollment to job readiness. It combines structured courses, assessments, certification, coding practice, contests, career roadmaps, resume building, jobs and internships, AI assistance, and administrative management.

The project follows a client-server architecture in which the React frontend communicates with a Spring Boot backend through REST APIs. The backend manages authentication, business logic and database operations, while MySQL provides persistent storage. Google Gemini is integrated to provide AI-assisted support.

2. Project Objectives

- Provide a secure role-based platform for students and administrators.
- Deliver structured courses through a Course → Module → Lesson learning model. □ Track student enrollment, lesson completion and overall learning progress.
- Assess student knowledge and provide verifiable certificates after successful completion.
- Develop technical skills through coding practice, contests, badges and leaderboards.
- Guide students using role-based career roadmaps and weekly study plans.
- Provide career-readiness tools through an ATS-enabled resume builder and jobs board.
- Provide an AI Assistant for conversational learning and platform support.
- Provide administrators with user, course, certificate, analytics and support management.

3. Project Scope

The scope is divided into five major areas to keep development organized.

Area	Scope
Learning	Courses, modules, lessons, enrollment, progress, notes, bookmarks, wishlist, reviews, quizzes and assessments.
Skill Validation	Coding practice, contests, automated test cases, leaderboards, badges and certificates.
Career Readiness	Career Roadmap, weekly study plans, Resume Builder, ATS score, Jobs & Internships and profile.
AI Support	Gemini-powered conversational assistant for learning and platform guidance.
Administration	User/course/certificate management, analytics, support tickets, audit logs and report exports.

4. Functional Module Plan

4.1 Landing Page

Introduce the platform, communicate its learning and career benefits, and provide clear navigation to registration and login.

4.2 Authentication and Security

Implement email/password registration and login, Google OAuth2, JWT-based stateless authentication, Student/Admin roles, password reset, OTP verification and BCrypt password hashing.

4.3 Student Dashboard and Progress

Display enrolled courses, completion percentage, completed courses, active courses, learning statistics and progress charts, with quick access to major student features.

4.4 Courses and Lessons

Provide searchable courses with a Course → Module → Lesson hierarchy, enrollment, lesson completion, notes, bookmarks, wishlist and reviews.

4.5 Assessments and Certification

Provide course assessments, result processing and certificate eligibility. Successful students receive unique, verifiable certificates with QR-based verification.

4.6 Coding Practice

Provide coding questions, an in-browser editor, automated test-case submission, difficulty-based progress, badges and a global leaderboard.

4.7 Contests

Provide contest registration, timed participation, submissions, live rankings, discussions and contestspecific certificates.

4.8 Career Roadmap

Provide role-based roadmaps for Full Stack Developer, Java Backend Developer, Frontend Developer, Python Developer and Cloud Engineer, organized as a 12-week curriculum.

4.9 Resume Builder and Jobs

Allow students to build resumes using education, skills, projects, experience and achievements, view an ATS score, export a PDF and explore job/internship opportunities.

4.10 AI Assistant

Integrate Google Gemini through the backend to answer learning and platform-related questions in a conversational interface.

4.11 Admin Dashboard

Provide analytics, user and course management, certificate management, revenue/enrollment insights, support-ticket resolution, audit logs and report exports.

5. Technology Plan

Layer	Technology	Purpose
Frontend	React + Vite + Tailwind CSS	Responsive user interface and page navigation
Visualization	Recharts	Student and admin analytics
Backend	Spring Boot	REST APIs and business logic
Security	Spring Security + JWT + OAuth2	Authentication and authorization
Persistence	JPA/Hibernate + MySQL	Relational data management
AI	Google Gemini API	AI Assistant and learning support
Email	Spring Mail	OTP and password-reset workflows
Documents	OpenPDF / frontend PDF tools	Certificates and resume export

6. Development Strategy

Development will be carried out incrementally. Each major phase will produce a usable deliverable before the next phase begins. Frontend, backend and database work will be integrated continuously rather than being completed as isolated components.

Phase A — Planning and Design

1. Finalize project objectives, scope, users and functional requirements.
2. Define the system architecture and major modules.
3. Design the MySQL entity structure and relationships.
4. Prepare the REST API and security approach.
5. Finalize UI structure and navigation.

Phase B — Core Development

6. Build the landing page and common frontend components.
7. Implement registration, login, JWT and Google OAuth2.
8. Develop courses, modules, lessons, enrollment and progress tracking.
9. Implement assessments, results and certificate generation.
10. Develop the admin management foundation.

Phase C — Advanced Features

11. Develop coding practice and contest functionality.
12. Implement the Career Roadmap and weekly study plans.
13. Develop Resume Builder, ATS scoring and Jobs & Internships.
14. Integrate the Google Gemini AI Assistant.
15. Complete notifications and support-ticket functionality.

Phase D — Integration and Quality

16. Perform API, functional, integration and security testing.
17. Verify student and admin end-to-end workflows.
18. Fix defects and improve usability.
19. Prepare deployment, documentation and final demonstration.

7. Work Breakdown Structure

WBS	Work Package	Main Activities	Deliverable
1.0	Planning	Requirements, scope, roles, repository	Approved project plan
2.0	Architecture	System, API, security and DB design	Architecture specification
3.0	UI/UX	Landing, auth, dashboard and navigation	Frontend foundation
4.0	Learning	Courses, lessons, enrollment, progress	Learning module
5.0	Assessment	Quizzes, assessments, results	Assessment module
6.0	Certification	Certificate, QR and verification	Credential module
7.0	Skill Practice	Coding, contests, badges, leaderboard	Practice and contest module
8.0	Career	Roadmap, resume, ATS, jobs	Career module
9.0	AI & Support	Gemini Assistant, tickets, notifications	Support services
10.0	Admin	Users, courses, analytics, reports	Admin dashboard
11.0	Testing	Functional, integration, security	Validated system
12.0	Submission	Deployment, report, presentation	Final submission package

8. Twelve-Week Project Schedule

The schedule below is the primary execution plan. It is aligned with the 12-week activity sequence documented in the supplied project report, while presenting the work as a clear plan with planned outputs.

Week	Focus	Planned Work	Output
Week 1	Initiation & Requirements	Finalize objectives, scope, features, roles and repository.	Requirements + project plan
Week 2	Architecture & Database	Finalize client-server architecture, REST API plan, security and MySQL schema.	Architecture + DB design

Week 3	UI Foundation	Build landing page, common components and database foundation.	Landing page + base UI
Week 4	Authentication	Implement registration, login, JWT, Google OAuth2 and role-based access.	Secure authentication
Week 5	Learning Module	Build Course → Module → Lesson hierarchy, enrollment and progress.	Working learning flow
Week 6	Assessment & Certificate	Implement assessments, results, certificate generation, QR and verification.	Assessment + certificate
Week 7	Admin Foundation	Build user, course and certificate management and initial analytics.	Admin management
Week 8	Progress Analytics	Implement student statistics, charts and course-wise progress.	Progress dashboard
Week 9	Coding Practice	Build coding questions, editor, test cases, progress, badges and leaderboard.	Coding arena
Week 10	Contests & Roadmap	Build contests, rankings and role-based 12-week Career Roadmaps.	Contest + roadmap
Week 11	Career Tools	Build Resume Builder, ATS score, jobs, profile, settings and notifications.	Career-readiness tools
Week 12	AI, Support & Final QA	Integrate Gemini Assistant, support, complete testing and final documentation.	Final integrated system

9. Key Milestones

Milestone	Date	Completion Target
Project Kickoff	29 June 2026	Objectives, scope, architecture direction and responsibilities finalized.
Prototype / First Draft	08 July 2026	Initial Spring Boot backend, JWT authentication and core UI available.
Mid-Term Review	23 July 2026	Course/assessment/certificate flow and admin foundation reviewed.
Feature Completion	13 August 2026	Career Roadmap, Coding Practice, Contests, Resume, Jobs and AI Assistant completed.
Final Presentation	01–02 September 2026	Working system demonstrated through major end-to-end workflows.

10. Planned Review Points

- End of Week 2: Review architecture, database and API design.
- End of Week 4: Review authentication and base frontend.
- End of Week 6: Review learning, assessment and certification flow.
- End of Week 8: Review dashboards and analytics.
- End of Week 10: Review coding, contests and career roadmap.
- End of Week 12: Final integration, testing and submission review.

11. Team Responsibility Plan

Responsibilities are divided by the major technical areas documented in the project report. Team members will coordinate during integration and testing so that the final system functions as one application.

Member	Primary Area	Responsibilities
Kartik Saini	Frontend + AI	Frontend development, UI integration and Google Gemini AI Assistant integration.
Harshit Raj	Frontend	Student-facing screens, responsive UI and frontend feature integration.
S. Kavipriya	Backend	Spring Boot APIs, business logic, authentication/security and backend integration.
Gayathri	Database	MySQL schema, relationships, database operations and database integration.
Mentor: Shakthi Gopala Krishnan	Mentoring	Technical guidance, review, feedback and project direction.

12. Testing and Quality Plan

Testing Type	What Will Be Verified
Unit Testing	Backend services, controllers, utility functions and important frontend components.
API Testing	REST endpoints, validation, authentication, authorization and error responses.
Integration Testing	React ↔ Spring Boot ↔ MySQL and external service communication.
Functional Testing	Registration, enrollment, lessons, assessment, certificate, coding, contests, roadmap, resume, jobs and admin workflows.
Security Testing	JWT, OAuth2, role restrictions, password security and protected routes.
Usability Testing	Navigation, responsive layout, loading states, errors and overall user experience.
Performance Testing	Dashboard, APIs and reports under representative data.
Acceptance Testing	Complete Student and Admin workflows before final submission.

13. Risk Management

Risk	Impact	Planned Mitigation
Authentication complexity	Login or Google Sign-In errors.	Test JWT and OAuth2 independently and together.
Course hierarchy complexity	Incorrect progress or enrollment data.	Use clear entity relationships and test progress calculations.
Certificate issues	Duplicate or invalid credentials.	Use unique certificate IDs and controlled generation logic.
Coding evaluation	Incorrect submission or ranking results.	Validate test cases and perform repeated integration testing.
Gemini/API failure	AI Assistant temporarily unavailable.	Use graceful error handling and retry/fallback messaging.

Database consistency	Incorrect records or analytics.	Use validation, constraints, transactions and backups.
Schedule changes	Development delay.	Prioritize core features and freeze major scope before final QA.
Deployment problems	Demo environment failure.	Test deployment configuration before final presentation.

14. Resources Required

- Node.js 18+ for frontend development.
- Java 17+ and Maven for Spring Boot development.
- MySQL 8+ for database storage.
- Git and GitHub for source-code management.
- Google OAuth2 credentials for Google Sign-In.
- Google Gemini API access for AI functionality.
- VS Code or an equivalent development environment.
- Modern web browsers for functional and responsive testing.
- Deployment environment for the final demonstration.

15. Final Deliverables

20. Working Enterprise Learning Platform web application.
21. Landing page, signup and login pages.
22. Student dashboard with learning statistics.
23. Course, module, lesson and enrollment functionality.
24. Assessment and certificate-generation system.
25. Coding practice arena, contests, badges and leaderboards.
26. Role-based Career Roadmap with weekly learning plans.
27. ATS-enabled Resume Builder and PDF export.
28. Jobs & Internships board.
29. Google Gemini-powered AI Assistant.
30. Admin dashboard with management and analytics.
31. Support tickets, notifications and account settings.
32. MySQL database and Spring Boot REST API implementation.
33. Testing evidence, project documentation and final presentation.

16. Success Criteria

- Student and Admin users can securely authenticate and access the correct features.
- Students can enroll in courses and receive accurate learning-progress updates.
- Assessments produce correct results and successful students receive verifiable certificates.
- Coding submissions and contest rankings are recorded correctly.
- Career Roadmaps provide structured role-based learning plans.
- Students can create resumes and access job/internship opportunities.
- The AI Assistant provides useful platform and learning-related responses.
- Administrators can manage core entities and view required analytics.
- Frontend, backend, database and external services integrate reliably.
- The final application passes functional, integration and security testing.

17. Expected Outcome

The completed system will provide a single platform for learning, skill validation and career preparation. Students will be able to learn through structured courses, monitor progress, take assessments, earn certificates, practice technical skills, participate in contests, follow career roadmaps, build resumes and explore job opportunities. Administrators will be able to manage the platform and monitor its activities through analytics and management tools.

18. Conclusion

This project plan provides a clear development roadmap from project initiation to final evaluation. The plan separates the work into manageable phases, defines responsibilities, establishes weekly deliverables, identifies risks and specifies measurable success criteria. It is designed to support systematic implementation of the Enterprise Learning Platform with Skill and Career Guidance System while maintaining focus on learning quality, technical skill development and career readiness.

19. Reference Basis

This project plan is based on the supplied Infosys Springboard Virtual Internship 7.0 completion report for the Enterprise Learning Platform with Skill and Career Guidance System. The report documents the project's objectives, technology architecture, module descriptions, 12-week activity sequence, milestone dates, implementation details, challenges and learning outcomes.